PYTHON ACTIVITY-BASED LEARNING REPORT

Name: Thanuj Lanke	**Roll No:** 26000175
Branch: CSE	**Course:** Python Programming

Case 1: Progressive Electricity Billing

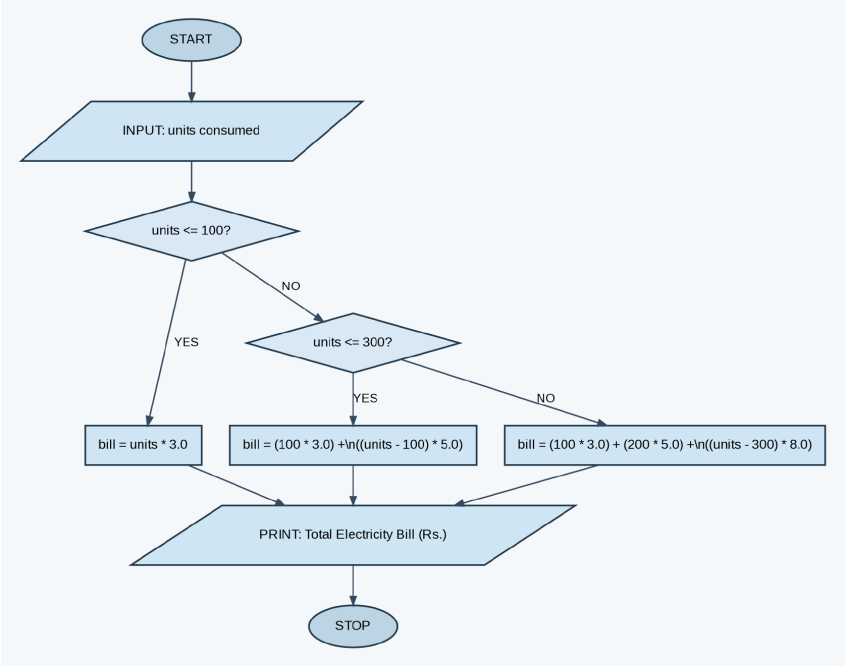
Problem Statement

Write a Python program to calculate the progressive electricity bill based on units consumed using tiered pricing rules.

Algorithm

- Step 1: Start.
- Step 2: Read units consumed from user.
- Step 3: If units ≤ 100, bill = units × 3.0.
- Step 4: Else if units ≤ 300, bill = (100 × 3.0) + ((units - 100) × 5.0).
- Step 5: Else bill = (100 × 3.0) + (200 × 5.0) + ((units - 300) × 8.0).
- Step 6: Print total bill. Stop.

Flowchart



Source Code & Output

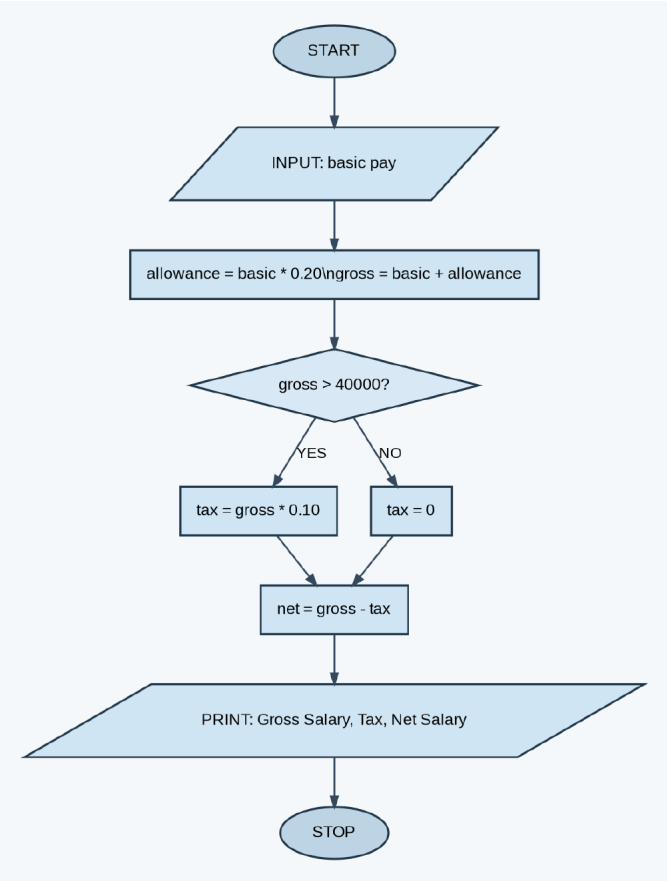
```
units = int(input("Enter units consumed: "))
if units <= 100:
    bill = units * 3.0
elif units <= 300:
    bill = (100 * 3.0) + ((units - 100) * 5.0)
else:
    bill = (100 * 3.0) + (200 * 5.0) + ((units - 300) * 8.0)
print("Total Electricity Bill: Rs.", bill)
```

```
Enter units consumed: 350 | Total Electricity Bill: Rs. 1700.0
```

Case 2: Employee Payroll Analysis

Problem Statement & Flowchart

Calculate gross salary, tax deductions based on income slab, and net salary for an employee.



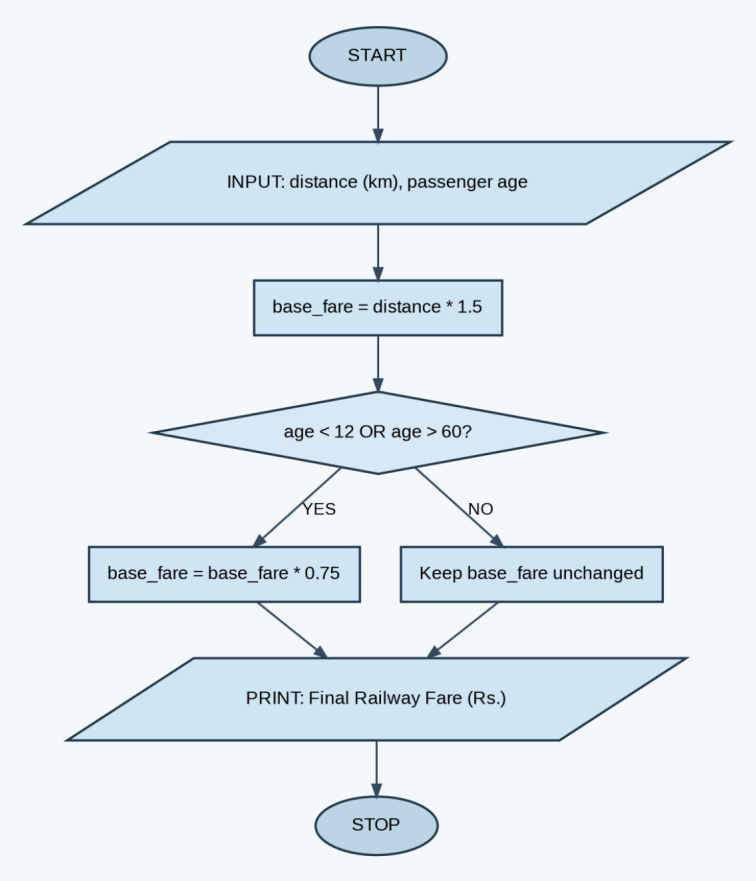
Source Code & Output

```
basic = float(input("Enter basic pay: "))
allowance = basic * 0.20
gross = basic + allowance
tax = gross * 0.10 if gross > 40000 else 0
net = gross - tax
print("Gross Salary:", gross, "Tax:", tax, "Net:", net)
```

```
Enter basic pay: 10000 | Gross Salary: 12000.0 | Tax: 0 | Net: 12000.0
```

Case 3: Railway Fare Calculator

Flowchart



Source Code & Output

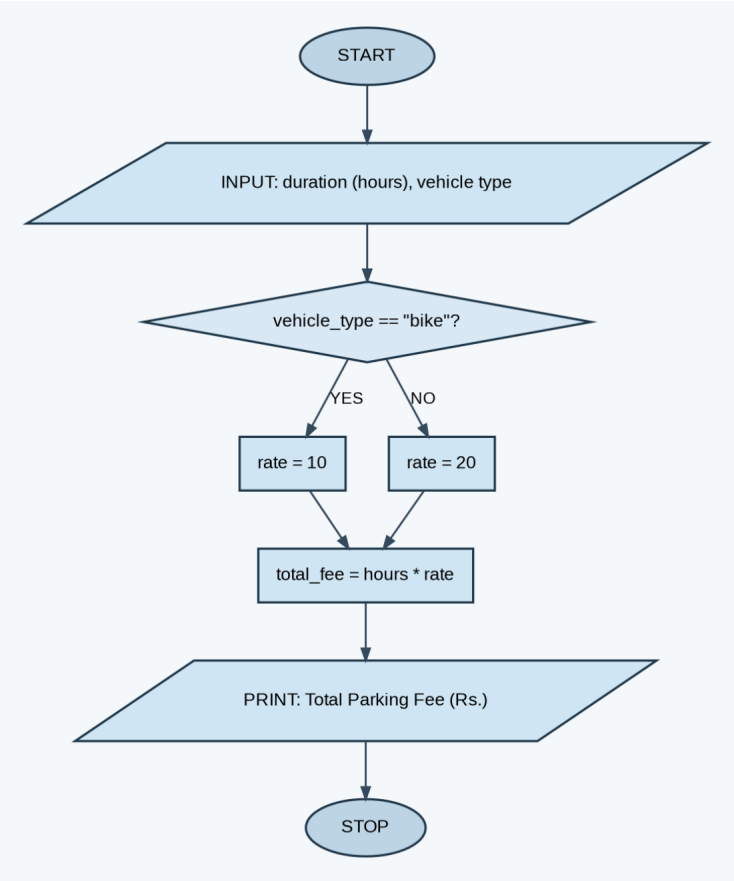
```
distance = int(input("Enter distance in km: "))
age = int(input("Enter passenger age: "))
base_fare = distance * 1.5
if age < 12 or age > 60:
    base_fare = base_fare * 0.75
print("Final Railway Fare: Rs.", base_fare)
```

```
Enter distance in km: 69 | Age: 19 | Final Railway Fare: Rs. 103.5
```

Case 4: Smart Parking Fee

Problem Statement & Flowchart

Calculate total parking fees depending on vehicle type (Bike or Car) and duration in hours.



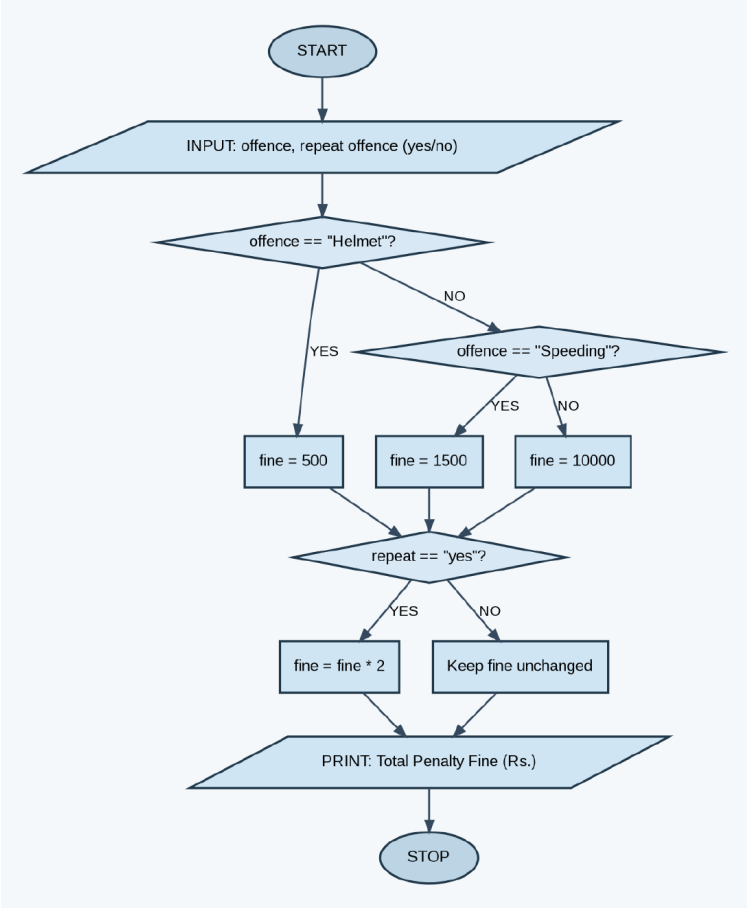
Source Code & Output

```
hours = int(input("Enter duration in hours: "))
vehicle_type = input("Enter vehicle type (Bike/Car): ").lower()
rate = 10 if vehicle_type == "bike" else 20
total_fee = hours * rate
print("Total Parking Fee: Rs.", total_fee)
```

```
Duration: 3 | Type: bike | Total Parking Fee: Rs. 30
```

Case 5: Traffic Violation Penalty

Flowchart



Source Code & Output

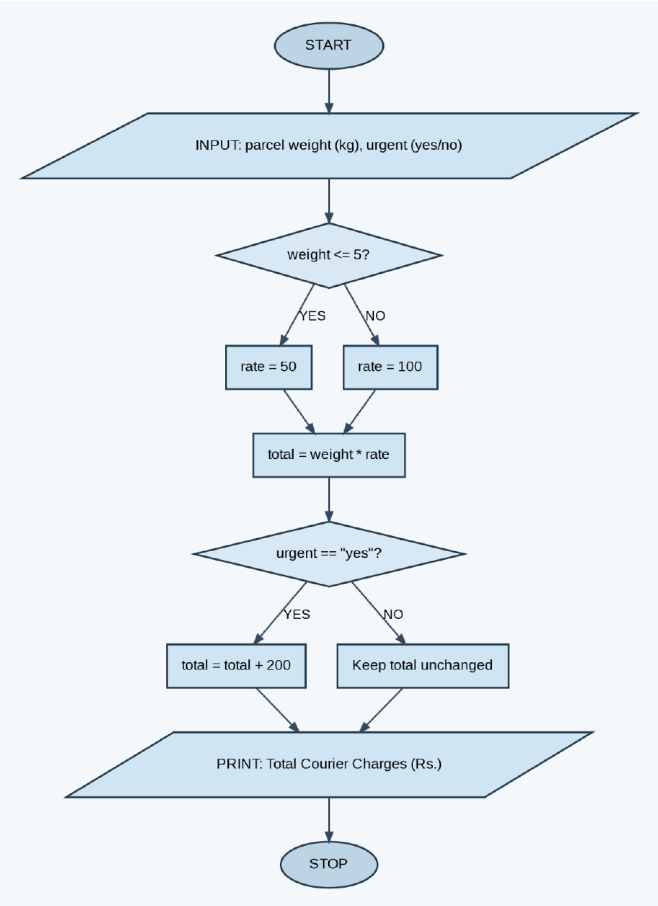
```
offence = input("Enter offence: ")
repeat = input("Is it a repeat offence? (yes/no): ")
fine = 500 if offence == "Helmet" else (1500 if offence == "Speeding" else 10000)
if repeat == "yes":
    fine *= 2
print("Total Penalty Fine: Rs.", fine)
```

```
Offence: Helmet | Repeat: no | Total Penalty Fine: Rs. 500
```

Case 6: Courier Charge Calculator

Problem Statement & Flowchart

Calculate courier charges based on parcel weight and urgency status.



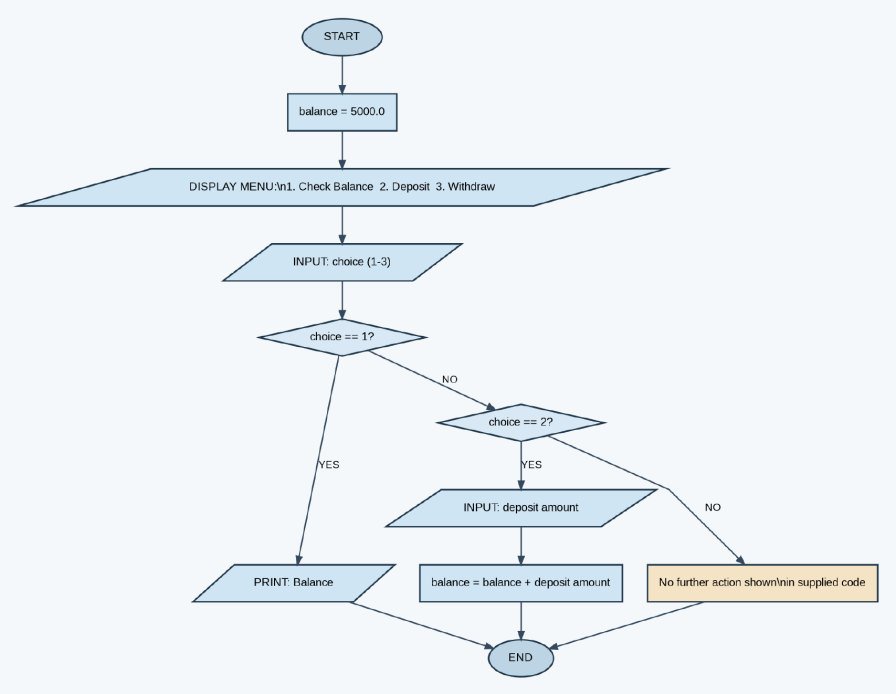
Source Code & Output

```
weight = float(input("Enter weight in kg: "))
urgent = input("Is it urgent? (yes/no): ")
rate = 50 if weight <= 5 else 100
total = weight * rate
if urgent == "yes":
    total += 200
print("Total Courier Charges: Rs.", total)
```

```
Weight: 30 | Urgent: yes | Total Courier Charges: Rs. 3200.0
```

Case 7: ATM Menu Simulation

Flowchart



Note: the supplied code shows only the check-balance and deposit branches; the withdraw branch is not present.

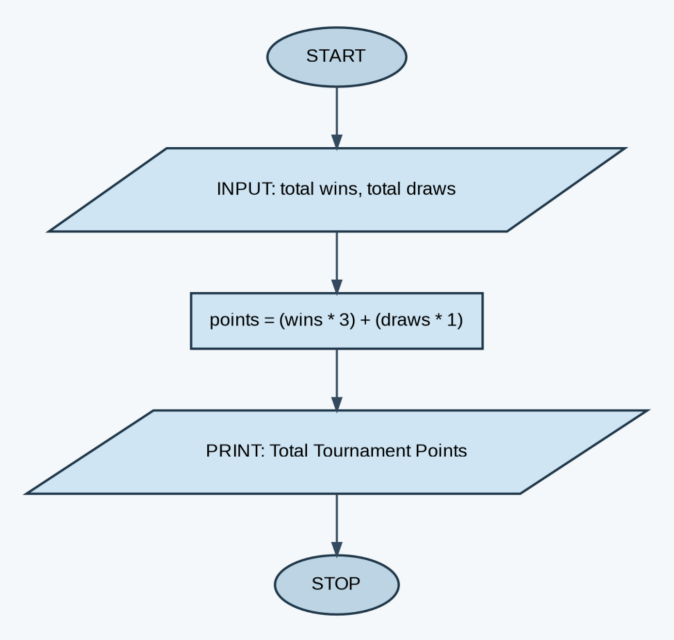
Source Code & Output

```
balance = 5000.0
print("1. Check Balance\n2. Deposit\n3. Withdraw")
choice = int(input("Enter choice (1-3): "))
if choice == 1:
    print("Balance:", balance)
elif choice == 2:
    balance += float(input("Deposit amount: "))
```

```
1. Check Balance | Choice: 1 | Balance: 5000.0
```

Case 8: Sports Tournament Points

Flowchart



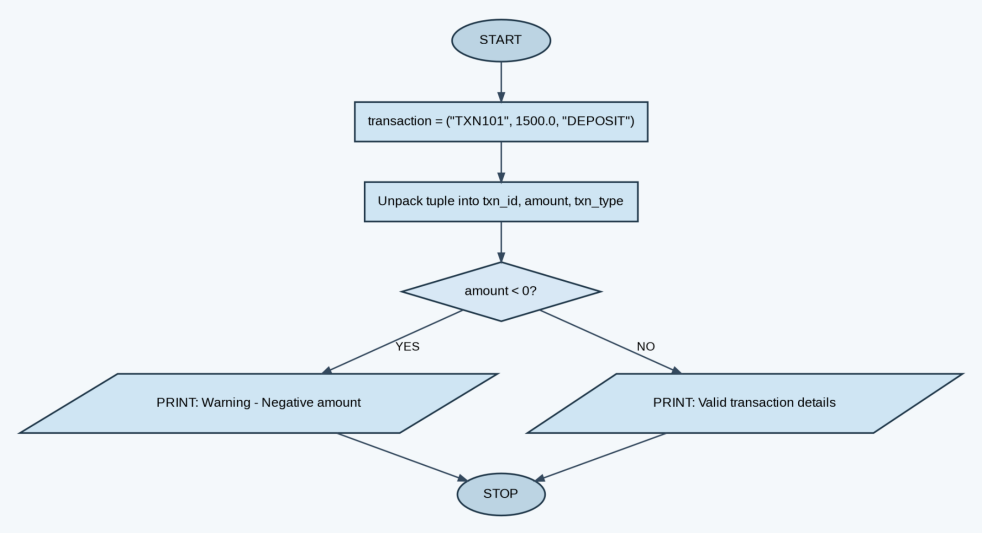
Source Code & Output

```
wins = int(input("Enter total matches won: "))
draws = int(input("Enter total matches drawn: "))
points = (wins * 3) + (draws * 1)
print("Total Tournament Points:", points)
```

```
Wins: 5 | Draws: 1 | Total Tournament Points: 16
```

Case 9: Transaction Audit using Tuples

Flowchart



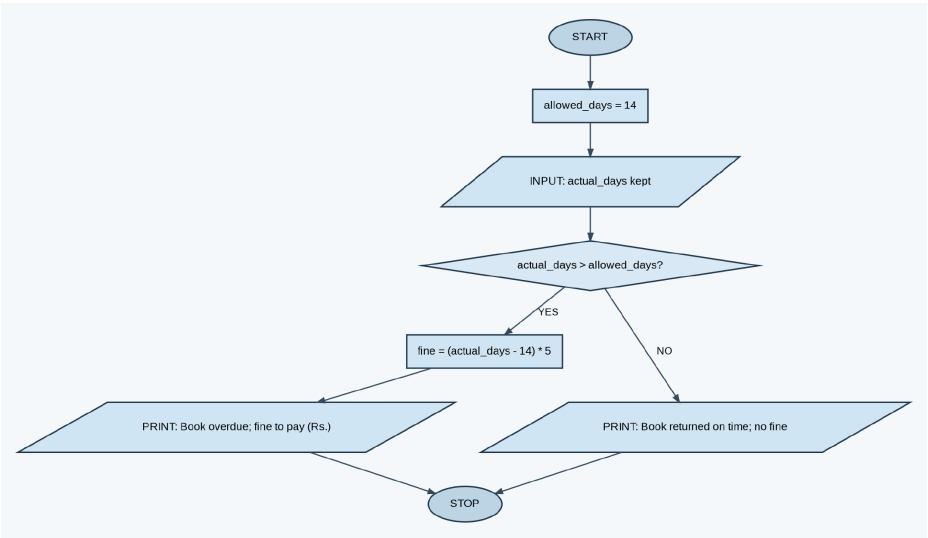
Source Code & Output

```
transaction = ("TXN101", 1500.0, "DEPOSIT")
txn_id, amount, txn_type = transaction
if amount < 0:
    print("Warning: Negative amount")
else:
    print(f"Valid {txn_type} of Rs.{amount} recorded.")
```

```
Valid DEPOSIT of Rs.1500.0 recorded for TXN101.
```

Case 10: Library Fine Calculator

Flowchart



Source Code & Output

```
allowed_days = 14
actual_days = int(input("Enter total days kept: "))
if actual_days > allowed_days:
    fine = (actual_days - allowed_days) * 5
    print("Book is overdue. Fine to pay: Rs.", fine)
else:
    print("Book returned on time. No fine!")
```

```
Enter total days kept: 30 | Book is overdue. Fine to pay: Rs. 80
```